Venkata Sri Sai Ramesh Murala

Machine Learning Engineer

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Professional Summary

Machine Learning Engineer with 4+ years of experience designing data-driven solutions across enterprise and technology environments. Strong expertise in machine learning algorithms, data processing, and model deployment, with a proven ability to improve predictive accuracy, streamline workflows, and deliver scalable solutions supporting business objectives and operational efficiency.

Core Skills

Machine Learning & AI: Supervised Learning, Unsupervised Learning, NLP, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Feature Engineering, Model Evaluation, Ensemble Methods

Generative AI & LLM Tools: LangChain, LlamaIndex, Prompt Engineering, Vector Databases (FAISS, Pinecone), Embedding Models, Semantic Retrieval

Programming & Libraries: Python, SQL, Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy

Cloud Platforms: AWS (SageMaker, S3, EC2, Lambda, Glue, API Gateway, CloudWatch)

MLOps & Deployment: Docker, Kubernetes, CI/CD Pipelines, Model Deployment, Model Monitoring, Experiment Tracking

Data Engineering & Processing: Data Cleaning, Data Transformation, ETL Pipelines, Data Modeling, Distributed Data Processing

API & Integration: REST APIs, Microservices Architecture, Real-Time Inference Pipelines, System Integration

Visualization & Reporting: Power BI, Tableau, Matplotlib, Data Storytelling

Core Competencies: Analytical Thinking, Problem Solving, Stakeholder Communication, Cross-Functional Collaboration

Professional Experience

Intuit | Machine Learning Engineer — Jan 2025 - Present

- Designed and deployed machine learning models using Python and AWS SageMaker, improving prediction accuracy for financial use cases and enabling data-driven decision-making across enterprise platforms by 34%.
- Developed natural language processing pipelines for text classification and sentiment analysis, improving customer insight extraction and supporting product enhancements across financial services applications by 29%.
- Implemented retrieval-augmented generation pipelines using vector databases and contextual retrieval, improving response relevance and reducing manual knowledge lookup time across internal systems by 36%.
- Processed large-scale datasets using AWS S3 and AWS Glue, enabling efficient data ingestion and transformation while reducing data processing time across analytics workflows by 32%.
- Built large language model–based solutions for document summarization and intelligent query handling, improving information accessibility and enhancing productivity across internal business teams by 31%.
- Deployed containerized machine learning services using Docker and AWS ECS, improving scalability and ensuring consistent performance across production environments supporting high-volume customer interactions.
- Integrated real-time inference pipelines using AWS Lambda and API Gateway, enabling low-latency predictions and improving responsiveness of machine learning-powered applications by 28%.
- Monitored model performance using evaluation metrics and AWS CloudWatch, identifying drift patterns and improving reliability and stability of predictions across production systems by 26%.

Tech Mahindra | Software Engineer — Nov 2021 - Jul 2023

- Developed backend services using Python and integrated data-driven functionalities, supporting application performance improvements and enabling scalable system operations across enterprise solutions by 30%.
- Designed and maintained APIs for seamless data exchange, improving system integration capabilities and ensuring consistent communication across distributed application environments.
- Assisted in implementing data processing pipelines, improving efficiency of data handling and enabling faster analysis across multiple application modules by 27%.
- Collaborated with cross-functional teams to understand business requirements and translate them into technical solutions aligned with project objectives and delivery timelines.
- Improved application performance through code optimization and efficient data handling techniques, reducing response times and enhancing user experience across platforms by 26%.
- Participated in testing and debugging activities, identifying issues and ensuring delivery of stable and reliable software solutions across development lifecycle phases.
- Maintained technical documentation and system workflows, supporting knowledge sharing and ensuring clarity across teams working on complex application environments.

Maruti Techlabs | Data Scientist Intern **May 2020 - Oct 2021**

- Developed machine learning models using Python to analyze business data, improving prediction accuracy and supporting data-driven decision-making across client projects by 32%.
- Performed data preprocessing and feature engineering, ensuring high-quality datasets for modeling and improving performance of machine learning algorithms across various use cases.
- Conducted statistical analysis and exploratory data analysis, identifying trends and patterns that supported business insights and improved reporting effectiveness by 28%.
- Built data visualization dashboards using Power BI, improving visibility into key metrics and enabling stakeholders to make informed decisions based on analytical insights.
- Collaborated with stakeholders to gather data requirements and deliver analytical solutions aligned with business objectives and project expectations.
- Evaluated model performance using validation techniques, ensuring accuracy and reliability of predictions across deployed machine learning solutions.
- Assisted in automating data workflows, reducing manual effort and improving efficiency across analytics and reporting processes by 25%.

Education

Master of Data Science | University of Memphis, Memphis, TN, USA

Projects

Enterprise Knowledge Assistant using RAG

- Designed a retrieval-augmented generation system integrating vector search and LLMs, improving response accuracy and reducing manual knowledge lookup time across internal business workflows by 35%.
- Implemented semantic search using embeddings and vector databases, enabling context-aware query handling and improving information retrieval efficiency across large document repositories.
- Deployed solution using AWS services, ensuring scalable data processing and enabling real-time responses for internal users across enterprise knowledge management systems.

Customer Sentiment Analysis using NLP

- Developed NLP models for sentiment classification using Python and machine learning techniques, improving customer feedback analysis accuracy and enabling data-driven product improvement decisions by 30%.
- Processed unstructured text data through preprocessing and feature engineering, improving model performance and ensuring reliable classification across large customer datasets.
- Visualized sentiment trends using dashboards, enabling stakeholders to monitor customer experience metrics and identify key areas for operational improvements.

Fraud Detection System using Machine Learning

- Built anomaly detection models to identify fraudulent transactions, improving detection accuracy and reducing financial risk exposure across transaction processing systems by 28%.
- Engineered features from transactional data, enabling improved pattern recognition and enhancing model performance across financial datasets.
- Deployed models using cloud-based infrastructure, ensuring scalability and supporting real-time fraud detection across high-volume financial systems.

Real-Time Recommendation System

- Designed recommendation engine using collaborative filtering techniques, improving user engagement and increasing conversion rates across digital platforms by 25%.
- Processed large-scale interaction data, enabling personalized recommendations and improving user experience across customer-facing applications.
- Integrated real-time inference pipelines, ensuring low-latency predictions and improving responsiveness of recommendation services.